

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

The command is "Show ip route ospf"

Part 4: Configure OSPF Passive Interfaces

g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129

Does R2 show up as an OSPF neighbor on R1? Yes

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you?

Mean the interface that directly connecting with R2 is either set to passive interface or link down, so R3 couldn't receive any Hello packet from it so it counted as link down. Also the route to 192.168.2.0/24 have the metric of 129, meaning it have go on a very long hop.

h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.
"router ospf 1" and "no passive-interface s0/0/1"

i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

The metric for 192.168.2.0/24 65 on R3, $100\text{Mbps} / 1.544\text{Mbps} = 64 + 1 \text{ Ethernet} = 65$
Is R2 listed as an OSPF neighbor to R3? Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

I want to change the OSPF default reference-bandwidth is because I want the OSPF metric to correctly reflect the real link speed and to find the shortest path to the destination in modern network.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

The cost of the 192.168.3.0/24 is calculated by reference bandwidth $100000 / 128 = 781$ then plus 1 total metric of 782. As the route to 192.168.23.0/30 using 1544 Kbps, overall metric is $781 + (100000 / 1544) = 781 + 64$ which is total 845 metric to reach the network 192.168.23.0.

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

Now the new accumulated cost to 192.168.23.0/24 is 1562, because R2 and R3 serial link have reduce to 128Kbps, therefore $781 + 781 = 1562$.

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

The R1 routing to the network 192.168.3.0/24 is now going through R2 is because we manually turn the link facing R3 which is S0/0/1 metric cost to higher, therefore OSPF will always choose the cumulated lowest metric route to be the best route.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

The router ID assignment is important when using the OSPF protocol because router ID is the important value of letting the router inside the same OSPF area to elect, if not strictly control the router ID, that might lead to operational confusion that hard to identify which router belong in the network topology, also instability if router ID is continually changing would lead to disrupt the traffic that affecting the routing table.

2. Why is the DR/BDR election process not a concern in this lab?

There is no DR/BDR is not the main concern inside this lab, is because all the router are directly connected to their neighbor, and mostly are point to point connection, therefore OSPF not using the DR/BDR election process due to each router have maximum of 2 router can be establish as a neighbor.

3. Why would you want to set an OSPF interface to passive?

We want to set a interface to passive is first because of safety, prevent OSPF Hello packet to send to untrusted network, like sending Hello packet to user network. Second is to higher the efficiency that setting the interface passive can let the router stop trying to form a neighbor in a impossible interface.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

The OSPF have the ability to separate different area, that make it the protocol of able to create a very scalable networks, that by separate different routing area to area number, that allowing limiting the scope of link-state advertisement .

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

The command is "network 10.0.0.0 0.255.255.255 area 0"

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

The router will use the EIGRP route because according to standard Administrative Distance(AD) that EIGRP is 90, OSPF is 110 and RIP is 120, because router will decide base on the AD instead of Metric first, therefore lower the AD then router will select it first.